Chapter 4. Countability and Separation Axioms

First and second countability, regular and normal spaces, Urysohn lemma, Urysohn metrization, Tietze extension

Contents

4.1 Countability axioms **2**

4.2 Separation axioms **2**

4.3 Normal spaces **2**

4.4 The Urysohn lemma **3**

4.5 The Urysohn metrization theorem **3**

4.6 The Tietze extension theorem **3**

4.7 Imbeddings of manifolds **4**

4.1 Countability axioms

X is **first countable** if each point has a countable neighbourhood basis, and **second countable** if the topology has a countable basis. A subset is **dense** if its closure is X; X is **separable** if it has a countable dense subset, and **Lindelöf** if every open cover has a countable subcover.

Theorem 4.1

In a first countable space, $x \in \overline{A}$ iff some sequence in A converges to x, and f is continuous iff it preserves convergent sequences. A second countable space is separable and Lindelöf. Subspaces and countable products of first (second) countable spaces are first (second) countable.

Metric spaces are first countable; for them, second countable, separable and Lindelöf are equivalent. $\mathbb{R}_\ell$ is first countable, separable and Lindelöf but not second countable; $\mathbb{R}_\ell^2$ (the Sorgenfrey plane) is not Lindelöf (the antidiagonal is closed and discrete), so products of Lindelöf spaces need not be Lindelöf.

4.2 Separation axioms

Assume one-point sets are closed. X is **regular** if a point and a disjoint closed set have disjoint neighbourhoods, and **normal** if two disjoint closed sets do. Normal $\Rightarrow$ regular $\Rightarrow$ Hausdorff.

Lemma 4.2

X is regular iff every neighbourhood U of x contains a neighbourhood V of x with $\overline{V} \subset U$; normal iff the same holds with closed sets A in place of points.

Subspaces and products of Hausdorff (regular) spaces are Hausdorff (regular); normality is not preserved: $\mathbb{R}_\ell$ is normal but $\mathbb{R}_\ell^2$ is not, and $S_\Omega \times \overline{S}_\Omega$ is a non-normal subspace of a normal space. $\mathbb{R}_K$ is Hausdorff but not regular.

4.3 Normal spaces

Theorem 4.3

(a) Regular second countable spaces are normal. (b) Metrizable spaces are normal. (c) Compact Hausdorff spaces are normal. (d) Well-ordered sets in the order topology are normal.

Proof. (b) For disjoint closed A, B, take $U = \bigcup_{a\in A} B(a, \epsilon_a/2)$, $V = \bigcup_{b\in B} B(b, \epsilon_b/2)$, where $B(a, \epsilon_a)$ misses B and $B(b, \epsilon_b)$ misses A; the triangle inequality makes $U \cap V = \emptyset$. (c) Apply Theorem 3.6(b) twice. (a) With a countable basis, choose basis sets U_n covering A with closures missing B and V_n covering B with closures missing A; then $U_n' = U_n - \bigcup_{k\le n} \overline{V_k}$ and $V_n' = V_n - \bigcup_{k\le n} \overline{U_k}$ have disjoint unions. ■

4.4 The Urysohn lemma

Theorem 4.4 (Urysohn lemma)

If X is normal and A, B are disjoint closed sets, there is a continuous $f : X \to [0,1]$ with $f = 0$ on A and $f = 1$ on B.

Proof. Enumerate the rationals P in $[0,1]$ with $1, 0$ first. Using Lemma 4.2 recursively, define open sets U_p ($p \in P$) with $A \subset U_0$, $U_1 = X - B$, and $\overline{U_p} \subset U_q$ whenever $p < q$: when U_r is to be defined, it lies between already-defined U_p and U_q with $p < r < q$ adjacent among those defined, and normality gives $\overline{U_p} \subset U_r \subset \overline{U_r} \subset U_q$. Extend by $U_p = \emptyset$ for $p < 0$ and $U_p = X$ for $p > 1$, and put $f(x) = \inf\{p : x \in U_p\}$. If $x \in \overline{U_r}$ then $f(x) \le r$, and if $x \notin U_r$ then $f(x) \ge r$; hence $f^{-1}((c,d))$ is open, as one checks for each x_0 by choosing rationals $c < p < f(x_0) < q < d$ and the neighbourhood $U_q - \overline{U_p}$. ∎

A space where points and closed sets are separated by continuous functions is **completely regular** (Tychonoff). Normal ⇒ completely regular ⇒ regular; subspaces and products of completely regular spaces are completely regular.

4.5 The Urysohn metrization theorem

Theorem 4.5 (Urysohn metrization)

Every regular second countable space is metrizable.

Proof. X is normal (Theorem 4.3(a)). For each pair of basis sets B_n, B_m with $\overline{B_n} \subset B_m$, Urysohn gives $g_{n,m} : X \to [0,1]$ equal to 1 on $\overline{B_n}$ and 0 off B_m. This countable family $\{f_n\}$ separates points from closed sets: for $x \in U$ open, some $f_n(x) > 0$ and $f_n = 0$ off U. Then $F(x) = (f_1(x), f_2(x), \dots)$ imbeds X in $[0,1]^\omega \subset \mathbb{R}^\omega$ (product topology), which is metrizable (Theorem 2.6). Injectivity and continuity are clear; F is open onto its image because $F(U) \supset F(X) \cap \pi_n^{-1}((0,\infty)) \ni F(x)$ for the f_n chosen above. ∎

The proof shows more generally (the **imbedding theorem**): a T_1 space with a family of continuous functions $X \to [0,1]$ separating points from closed sets imbeds in $[0,1]^J$; thus completely regular spaces are exactly the subspaces of compact Hausdorff spaces.

4.6 The Tietze extension theorem

Theorem 4.6 (Tietze extension)

If X is normal and $A \subset X$ is closed, every continuous $f : A \to [a,b]$ (resp. $\mathbb{R}$) extends to a continuous $g : X \to [a,b]$ (resp. $\mathbb{R}$).

Proof. For $f : A \to [-r, r]$, Urysohn gives g with $|g| \le r/3$ on X and $|f - g| \le 2r/3$ on A (separate $f^{-1}([-r, -r/3])$ and $f^{-1}([r/3, r])$). Iterating on the error gives g_n with $|g_n| \le \frac{1}{3}(\frac{2}{3})^{n-1} r$ and $|f - \sum_{k \le n} g_k| \le (\frac{2}{3})^n r$ on A; $\sum g_n$ converges uniformly to an extension. For $\mathbb{R}$-valued f compose with a homeomorphism $\mathbb{R} \cong (-1,1)$ and correct the extension where it reaches ± 1 by multiplying with an Urysohn function that vanishes on the closed set where $|g| = 1$ and equals 1 on A. ∎

Tietze's theorem in turn implies Urysohn's lemma, so it characterizes normality.

4.7 Imbeddings of manifolds

An m-**manifold** is a second countable Hausdorff space in which every point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^m$.

Theorem 4.7

Every compact m-manifold imbeds in $\mathbb{R}^N$ for some N.

Proof. Cover X by finitely many open sets U_i with imbeddings $g_i : U_i \to \mathbb{R}^m$, and choose a **partition of unity** ϕ_i dominated by $\{U_i\}$ (continuous, $\sum \phi_i = 1$, $\operatorname{supp} \phi_i \subset U_i$; it exists in normal spaces by the shrinking lemma and Urysohn). Then $F = (\phi_1, \ldots, \phi_n, \phi_1 g_1, \ldots, \phi_n g_n)$ is an injective continuous map of a compact space into $\mathbb{R}^{n(m+1)}$, hence an imbedding. ■